10 CSTRs v2

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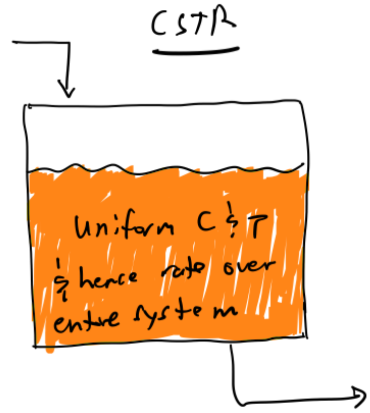

A key characteristic of the CSTR is it is well mixed, meaning the concentration and temperature are uniform over the entire volume

Specifically, CSTRs operate at
the exit concentration & temperature

Hence, the entire volume operates at one single rate, associated w/ the exit conditions

This phenomenon requires perfect mixing ; we will show later in the class that this is a large over-simplification that is rarely true in reality

CSTR mole balance:

- In terms of number of moles (N): $V = \frac{F_{A0} - F_A}{-r_A}$. In words, the reactor volume is equal to the amount of species A reacted in the reactor divided by the rate of consumption of species A.
  - If species A is a reactant, $F_{A0} - F_A$ is positive, and $-r_A$ is positive
  - If species A is a product, $F_{A0} - F_A$ is negative, and $-r_A$ is negative
  - If the "amount reacted" is larger, the volume needed is larger
  - If the rate of consumption is larger, the volume needed is smalle

- In terms of conversion (X): $V = \frac{F_{A0}X_A}{-r_A}$

- Same as with batch reactors, if you are solving a reactor design problem for a prototypical CSTR, you can just write down the CSTR mole balance without re-deriving it each time, with the following exceptions.
    - The problem statement asks you to derive the balance.
    - The reactor is not a prototypical CSTR.

- Let's do an example: The gas phase reaction 2A ↔ B exhibits elementary kinetics. It is carried out in a CSTR to 95% of the equilibrium conversion with PA0 = 10 atm and T = 400K. KC = 40 L/mol, kf = 0.1 L/mol/s, and $\upsilon_0 =$ 1 L/s. Determine the volume of the CSTR.

Step ① mole balance: $V = \frac{F_{A0}X}{-r_A}$

Step ② rate law: $-r_A = k_f\left(C_A^2 - \frac{C_B}{K_C}\right)$

Step ③ concentration equations; gas phase so

$$C_j = \frac{C_{LR,0}\left(\Theta_j - \frac{\nu_j}{\nu_{LR}}X\right)}{1+\varepsilon X} \qquad \varepsilon = y_{A0}\delta, \quad y_{A0} = 1, \quad \delta = \frac{1-2}{2} = -0.5$$

$$C_A = \frac{C_{A0}(1-X)}{1-0.5X} \qquad C_{A0} = \frac{P_{A0}}{RT} = 0.304\ \frac{mol}{L}$$

$$C_B = \frac{\frac{1}{2}C_{A0}X}{1-0.5X}$$

Step 4 what is X?

need $X = 0.95 X_e$. Use K_c to get X_e:

$$K_c = 40 = \frac{C_B}{C_A^2} = \frac{\frac{1}{2}C_{A0}X_e}{1-0.5X}\,\frac{(1-0.5X)^2}{C_{A0}^2(1-X)^2} = 0.86$$

$$0.95(0.86) = 0.82$$

Step 5 solve

0.304 $\frac{mol}{L}$ 1 L/s 0.82

$$V = \frac{F_{A0}X}{-r_A} = \frac{C_{A0}v_0 X}{k_f\left(C_A^2 - \frac{C_B}{K_C}\right)}$$

0.304 $\frac{mol}{L}$ 0.82

$$C_B = \frac{\frac{1}{2}C_{A0}X}{1-0.5X}$$

0.82

0.1 $\frac{L}{mol \cdot s}$ 40 $\frac{L}{mol}$

0.82

$$0.304\,\frac{mol}{L} \rightarrow \frac{C_{A0}(1-X)}{1-0.5X}$$

0.82

$$\boxed{V = 750\ L}$$